Network Analysis of Global Financial Markets

Project - 2 (MA47202) report submitted to the Indian Institute of Technology Kharagpur in partial fulfillment for the award of the degree of Bachelor of Science in Mathematics and Computing

by

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Under the supervision of

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Department of Mathematics

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Spring Semester, 2024-25

April 25, 2025

DECLARATION

I certify that

A. The work contained in this report has been done by me under the guidance of my supervisor.

B. The work has not been submitted to any other Institute for any degree or diploma.

C. I have conformed to the norms and guidelines given in the Ethical Code of Conduct of the Institute.

D. Whenever I have used materials (data, theoretical analysis, figures, and text) from other sources, I have given due credit to them by citing them in the text of the thesis and giving their details in the references. Further, I have taken permission from the copyright owners of the sources, whenever necessary.

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Place: Kharagpur (21MA10013)

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CERTIFICATE

This is to certify that the project report entitled “Network Analysis of Global Financial Markets” submitted by Aryan Garg (Roll No. 21MA10013) to Indian Institute of Technology Kharagpur towards partial fulfillment of requirements for the award of degree of Bachelor of Science in Mathematics and Computing is a record of bona fide work carried out by him under my supervision and guidance during Spring Semester, 2024-25.

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Date: April 25, 2025
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ABSTRACT

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Thesis supervisor: **Prof. Somesh Kumar**

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This thesis focuses on using a comprehensive network-topology framework to analyse financial markets across Indian, U.S., and global stock markets under multiple occasions - including the 2008 financial crisis, COVID-19 shock, Russia–Ukraine war, and recent trade-tariff skirmishes. This project analyses daily log-returns of 58-59 large-cap Indian and U.S. stocks (2006–2024) alongside 60 global stock indices (2018–2024) via rolling two-week shifts of a six-month window. Threshold networks (pairwise correlation > Threshold (θ)) yield snapshots from which average correlation, density, clustering coefficient, and characteristic path length are calculated. This study ranks the most influential stocks using five graph centralities - degree, closeness, betweenness, eigenvector, PageRank. Construction of minimum spanning trees (MSTs) helps us track network synchronization during the 2008 crash and COVID lockdowns, the 2022 Russia–Ukraine shock, and trade-tariff skirmishes. Post-shock topological shifts - such as India's migration between regional clusters and as well as migration of Russia and Ukraine during the war period - reveal evolving geopolitical and policy impacts.

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1 Introduction

Financial markets are inherently complex systems in which thousands of individual assets interact, and shocks in one region often propagate to other regions using economic drivers, trade, and capital flow. Traditional econometric models - factor analyses or pairwise correlation - often fall short of capturing the higher-order, complex relationships that take shape during periods of stress. In particular, crises such as the 2008 global financial crisis or the COVID-19 pandemic and the Russo-Ukrainian War reveal sudden, widespread contagion and cannot be easily explained or predicted using simple, static models. To solve this gap, network-topology methods have emerged as a powerful toolkit, involving graph-theoretic constructs with time-series analysis to conclude how markets merge into tight clusters or break into multiple different groups under different conditions.

This thesis represents markets - whether individual stocks or aggregated indices - as nodes in a graph, and encoding their correlation among them as edges, one can analyze three critical dimensions of systemic behavior:

- *Synchronization:* The total MST weight of the minimum spanning tree (MST) extracted from a connected graph offers a concise synchronization index, lower weight indicating greater overall correlation..
- *Systemic importance:* Calculation of multiple graph centrality measures (degree, closeness, betweenness, eigenvector, PageRank) identifies the most influential stocks (nodes) whose volatility amplifies changes in the graph.
- *Cluster realignment:* Threshold networks - linking only those pairs whose correlation exceeds a chosen threshold parameter θ - lead to the revelation of the migration of clusters as external shocks unfold.

Fifty-nine large-cap Indian and fifty-nine large-cap U.S. stocks, covering January 2006 to January 2025 and sixty country-level indices representing major economies and regions, spanning January 2018 to January 2025, encompass four landmark stress periods - the 2008 financial crisis, the COVID-19 pandemic, Russo-Ukrainian War, and U.S.-China

trade-tariff skirmishes - allowing for direct comparisons of network behavior across distinct geopolitical and economic contexts.

By integrating these analyses, the thesis offers actionable insights on the impact of the financial crisis on global financial markets by observing the dynamics of these stock networks, and the calculation of some network properties can also help in portfolio diversification.

2 Literature Review

2.1 Social-Network Analysis in Finance

Social Network Analysis can be used for studying complex inter-relationships among various stocks and to understand the internal structure of the global financial markets. It uses the concept that financial assets are interconnected nodes and captures relationships between them. A network-based approach helps in analyzing large amounts of time-series stock market data by making it easier to visualize. Network representations of correlation matrices reveal sectoral clustering and hierarchical market structures, aiding both diversification.

2.2 Graph Theory and Topological Methods

Graph theory provides the tools - vertices, edges, and paths - to model complex systems. Topology is concerned with the arrangement of the constituent parts of an object or a network. It discusses those properties of a geometric object that remain unchanged on deformation; for example, the shortest path between two nodes of a graph is independent

of the size of the nodes or edges of the graph.

Analysis of large amounts of data by using linear matrix methods can become a tiresome task. Topological methods help in better visualization of the data and quantitative analysis of market architecture by modeling it in the form of graphs and networks.

2.3 Correlation Threshold Networks

Correlation is a measure of statistical association. The Pearson correlation coefficient can be used to measure the correlation between two sets of data.

$$r = \frac{\sum (x - \bar{x})(y - \bar{y})}{\sqrt{\sum (x - \bar{x})^2 \sum (y - \bar{y})^2}}$$

The correlation matrix can be used for constructing adjacency matrices. A simple model for doing this is to employ threshold networks. A threshold network assigns an edge between two nodes by comparing the parameter based on which the graph is being constructed with a defined threshold (θ), filtering out weakly correlated nodes.

2.4 Network Properties

Some metrics and attributes can be calculated to analyze the properties and characteristics of the networks.

- *Average Correlation:* It gives an estimate of the correlation between two random nodes.
- *Network Density:* The proportion of direct connections in a network relative to the total number possible
- *Clustering Coefficient:* A measure of the likelihood that two associates of a node are associates. A higher clustering coefficient indicates a greater cliquishness or

compactness.

- *Characteristic Path Length:* It is the average shortest path length between two nodes.

2.5 Minimum Spanning Tree

A minimum spanning tree (MST) is a subset of the edges of a connected, edge-weighted undirected graph that connects all the vertices together, without any cycles and with the minimum possible total edge weight. The total MST weight then serves as a synchronization index: lower weight signals higher average correlation across the network. In other words, it is a spanning tree whose sum of edge weights is as small as possible.

2.6 Graph Centrality Measures

In graph analytics, centrality is a very important concept in identifying important nodes in a graph. It is used to measure the importance (or "centrality," as in how "central" a node is in the graph) of various nodes in a graph. We have used five centrality measures:

- *Betweenness Centrality:* It defines and measures the importance of a node in a network based upon how many times it occurs in the shortest path between all pairs of nodes in a graph.
- *Closeness Centrality:* This metric defines the importance of a node in a graph as being measured by how close it is to all other nodes in the graph. For a node, it is defined as the sum of the geodesic distance between that node to all other nodes in the network.
- *Degree Centrality:* This metric defines the importance of a node in a graph as being measured based on its degree, i.e., the higher the degree of a node, the more important it is in a graph.

- *Eigenvector Centrality:* Eigenvector centrality measures the importance of a node in a graph as a function of the importance of its neighbors. If a node is connected to highly important nodes, it will have a higher eigenvector centrality score as compared to a node that is connected to less important nodes.
- *PageRank:* The PageRank algorithm outputs a probability distribution used to represent the likelihood that a person randomly clicking on links will arrive at any particular page.

3 Methodology

The first phase of this study focused on constructing and analyzing threshold-based networks for 58 Indian and 59 U.S. stocks over the period 2006–2025. For each annual window, daily log returns were computed and used to generate unweighted graphs by applying a correlation threshold. Key network properties - average correlation, density, clustering coefficient, and characteristic path length - were tracked across stress events as mentioned before, and five graph centralities (degree, closeness, betweenness, eigenvector, and PageRank) were calculated to rank the most influential stocks in each market. In the second phase, this framework was extended to a set of 60 country-level indices (2018–2025), where pairwise correlations were transformed into distances and distilled into minimum spanning trees (MSTs). The total MST weight, along with structural snapshots, highlighted regional cluster shifts.

3.1 Data Collection and Preprocessing

We collected the financial data of 58 large-cap Indian stocks, 59 large-cap U.S. stocks, and 60 global country indices from ‘yfinance’ - an open-source library. We considered the weighted price as the price to be considered for the analyses.

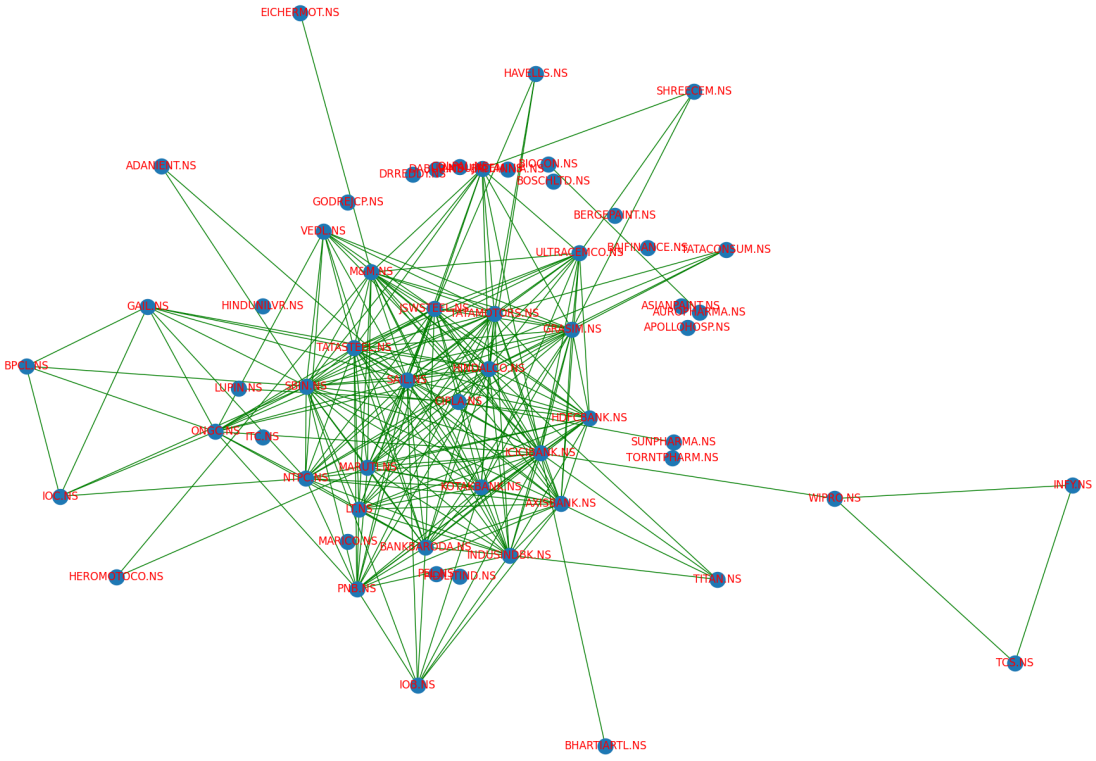
$$Weighted\ Price\ =\ (High\ +\ Low\ +\ Open\ +\ Close)/4$$

To avoid distortion in results due to fewer data points, the previous day's weighted price has been used in case of missing price data. For smoothing purposes, the daily logarithmic return has been considered.

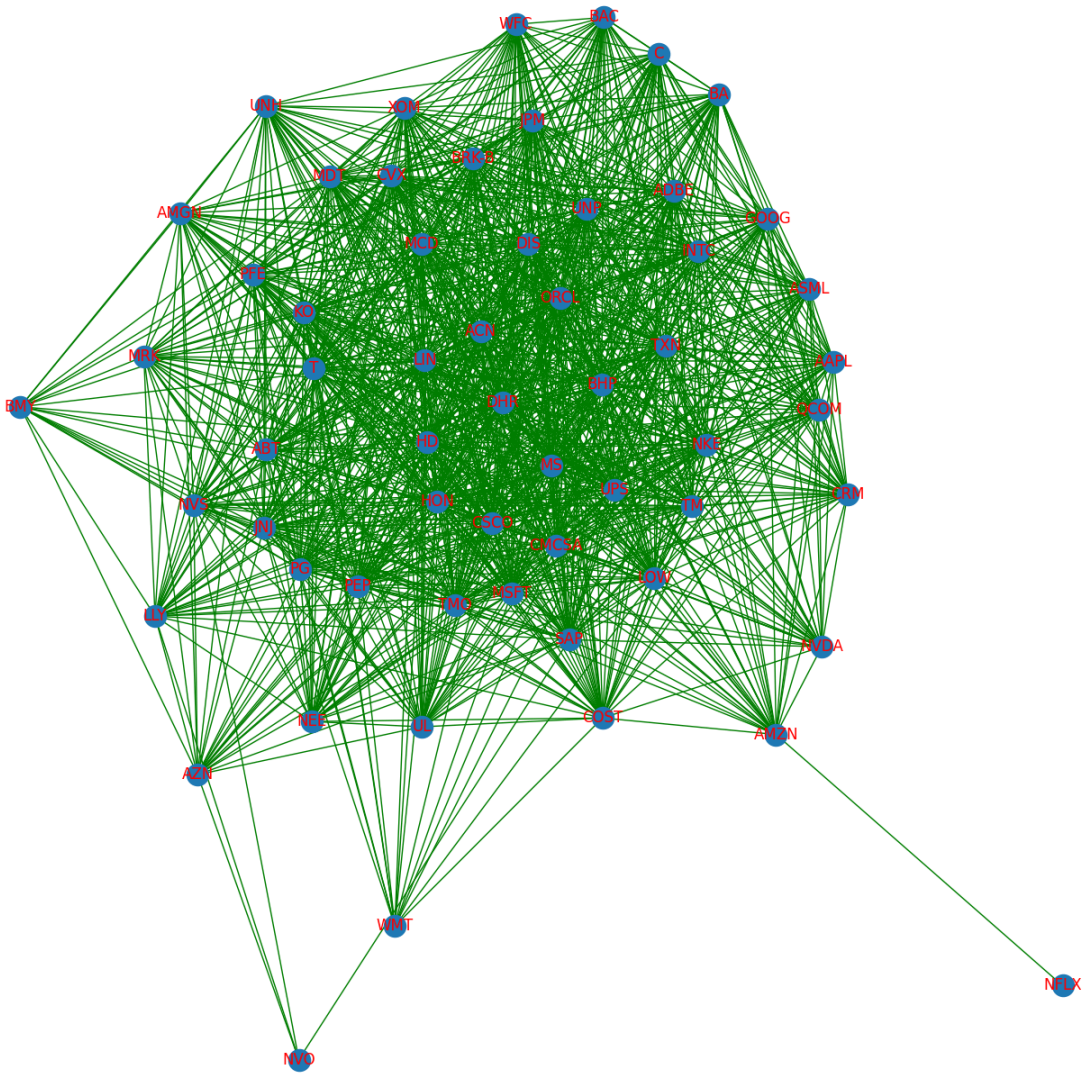
$$R_i(t)\ =\ ln\left[\ P_i(t)\ /\ P_i(t\ -\ 1)\right]$$

3.2 Network Construction

Correlation matrix is used for constructing adjacency matrices. A threshold parameter is fixed (0.36) for considering whether there is an edge between the node or not. This method is used for analysing India, U.S. stocks whereas while analysing global financial indices, correlation is directly used to form an adjacency matrix which is used for further analyses.



India Stock Network



U.S. Stock Network

For each time period of a year, a network of fully connected graphs of indices of India and the U.S. is constructed to see the clustering between different markets. This helps us to see the evolution of the clusters over the span of 19 years..

3.3 Topological Metrics and Graph Centralities

For each time period of a year, the mentioned network properties are calculated in order to see the effect of financial crises, natural pandemics, or economic and political wars on the Indian and U.S. stock markets. Peaks and dips in the plots of the network properties help us make inferences about the effect of multiple events.

With the help of a network graph, the mentioned graph centralities are calculated, and the nodes are ranked by the average of their percentile ranks of the five centralities. This helps us to find the most influential stocks in that region.

3.4 Constructing MST

A stock market index is a statistical measure that shows changes taking place in the stock market. Values of the underlying stocks are used to find the quantity of the stock market index.

We calculated the correlation matrix of global stock indices (for data from 2018 to 2025) and used it to form an MST. Minimum spanning trees provide better visual insights into the structural relationship of global stock markets by reducing complex networks to simpler ones.

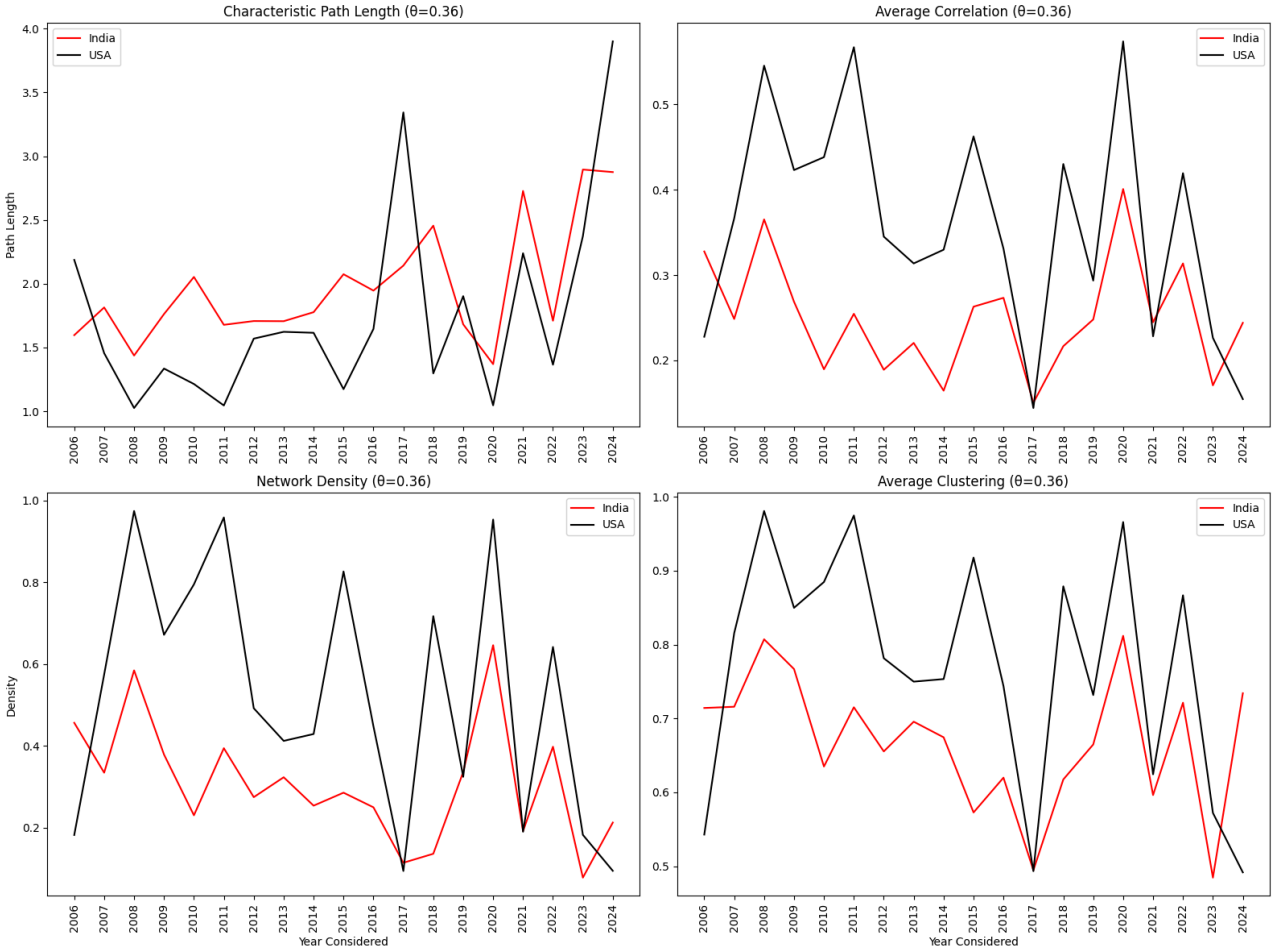
The weight of the MST is a good indicator of the level of synchronization in the global stock market. Changes in MST weight lead to market booms and crashes. MSTs are plotted for various time periods related to pre-COVID-19, during COVID-19, post-COVID-19, the Russia-Ukraine War, and trade tariff skirmishes in order to see the regional shifts between the countries during these times due to natural, economic, or political changes.

4 Results and Observations

4.1 Contrasting Network Properties

The four network properties showed contrasting natures over 19 years, from 2006 to 2024. During financial crises due to economic and political changes, markets become highly correlated with each other, which leads to booms in average correlation for that year. Due to high correlation, network density increases and hence reduces the characteristic path length and increases the clustering coefficient.

The following plot shows the booms and dips in the network properties:



Booms in plots of Average Correlation, Network Density, Average Clustering, and a fall in path length can be observed as expected during 2007-08 (Global Financial Crisis) and

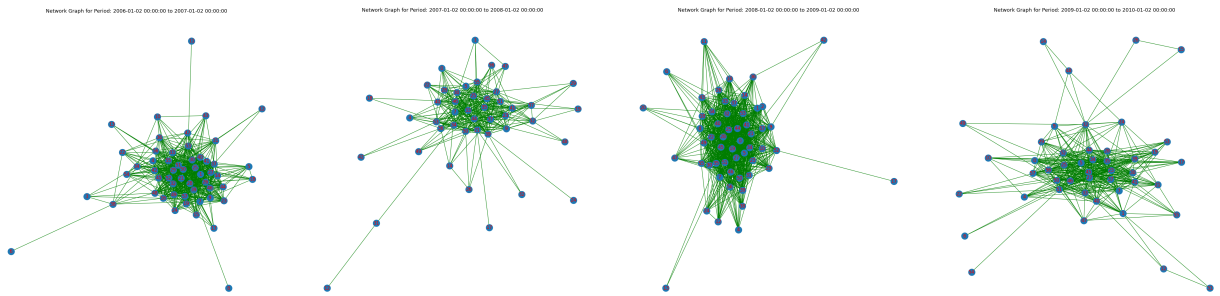
2019-20 (COVID-19) and 2022 (Russia-Ukraine War).

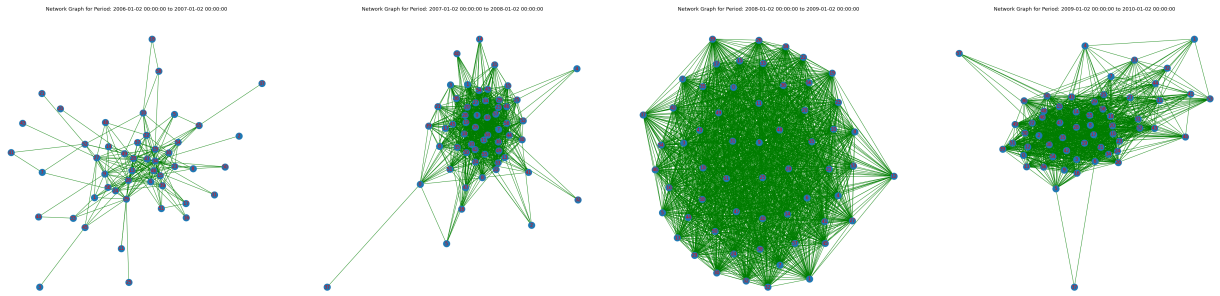
4.2 Stock Networks

Stock networks made from adjacency matrices give a correct idea related to the market growth or fall of the market of Indian and U.S. markets.

2006 2007 **2008** 2009

INDIA

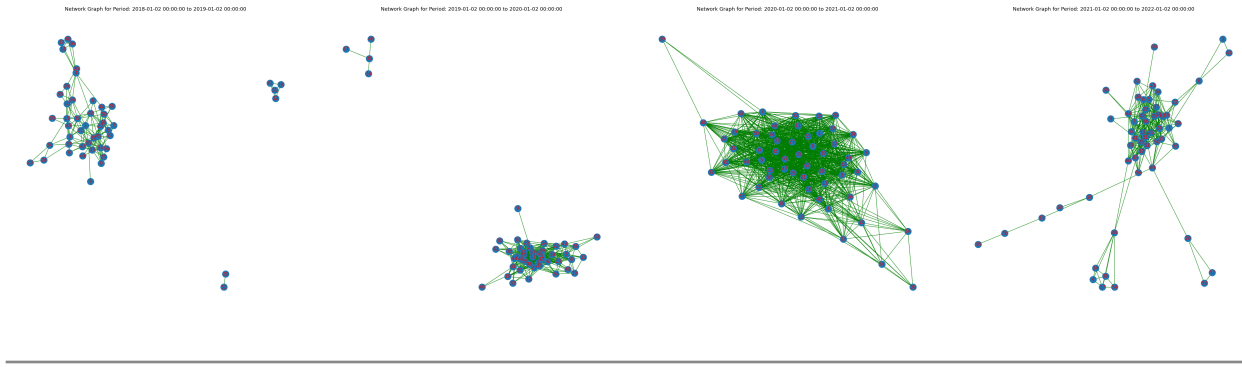


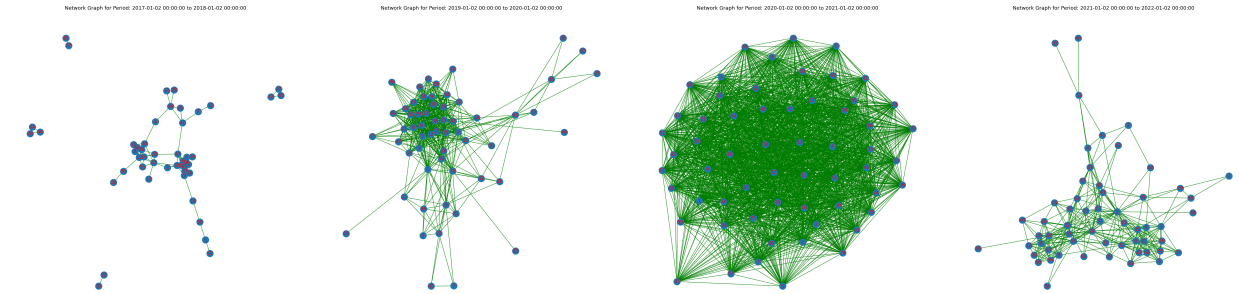


U.S.

2018 2019 **2020** 2021

INDIA

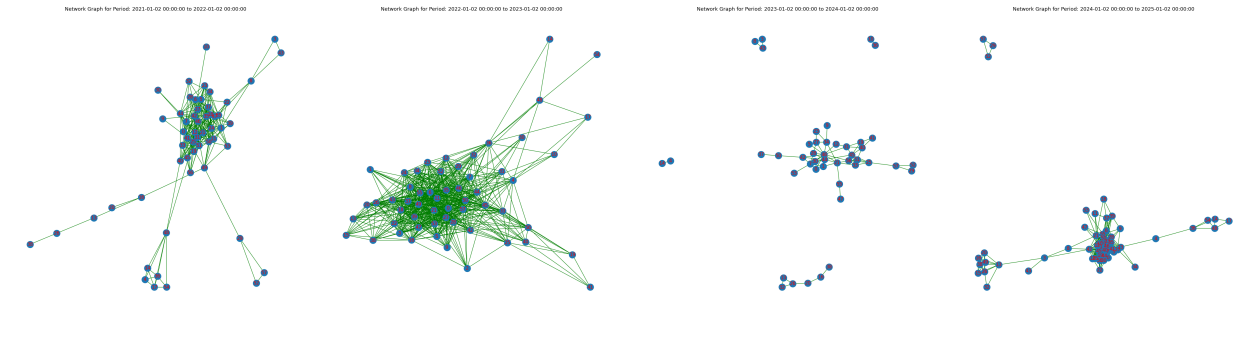


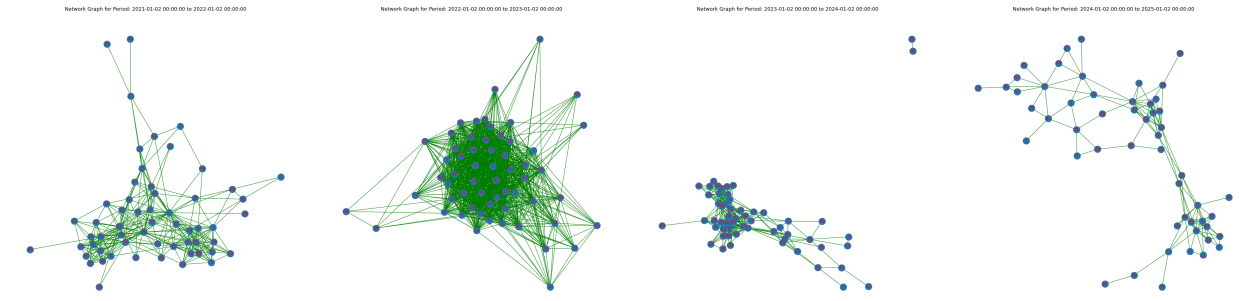


U.S.

2021 **2022** 2023 2024

INDIA





U.S.

It can be seen from the graphs that during an economic crisis, the stocks in a market become more correlated. Here, this can be seen for the 2008 financial crisis, the 2020 COVID-19 crisis, and the 2022 Russian-Ukrainian War. The graphs during these years are most dense for both India and the US. Whenever the stock market experiences growth, the markets become less correlated, as can be seen in the years 2018, 2021, and 2023.

On average, the U.S. stock market is larger than the Indian stock market, and it also comes to the point that the U.S. market is strongly correlated compared to the Indian market. An investor in India can diversify their portfolio in a better manner than one investing in the U.S. stock market. Uncorrelated clusters (sectors) form a healthy market and are easier for diversification.

4.3 Influential Stocks in India & USA

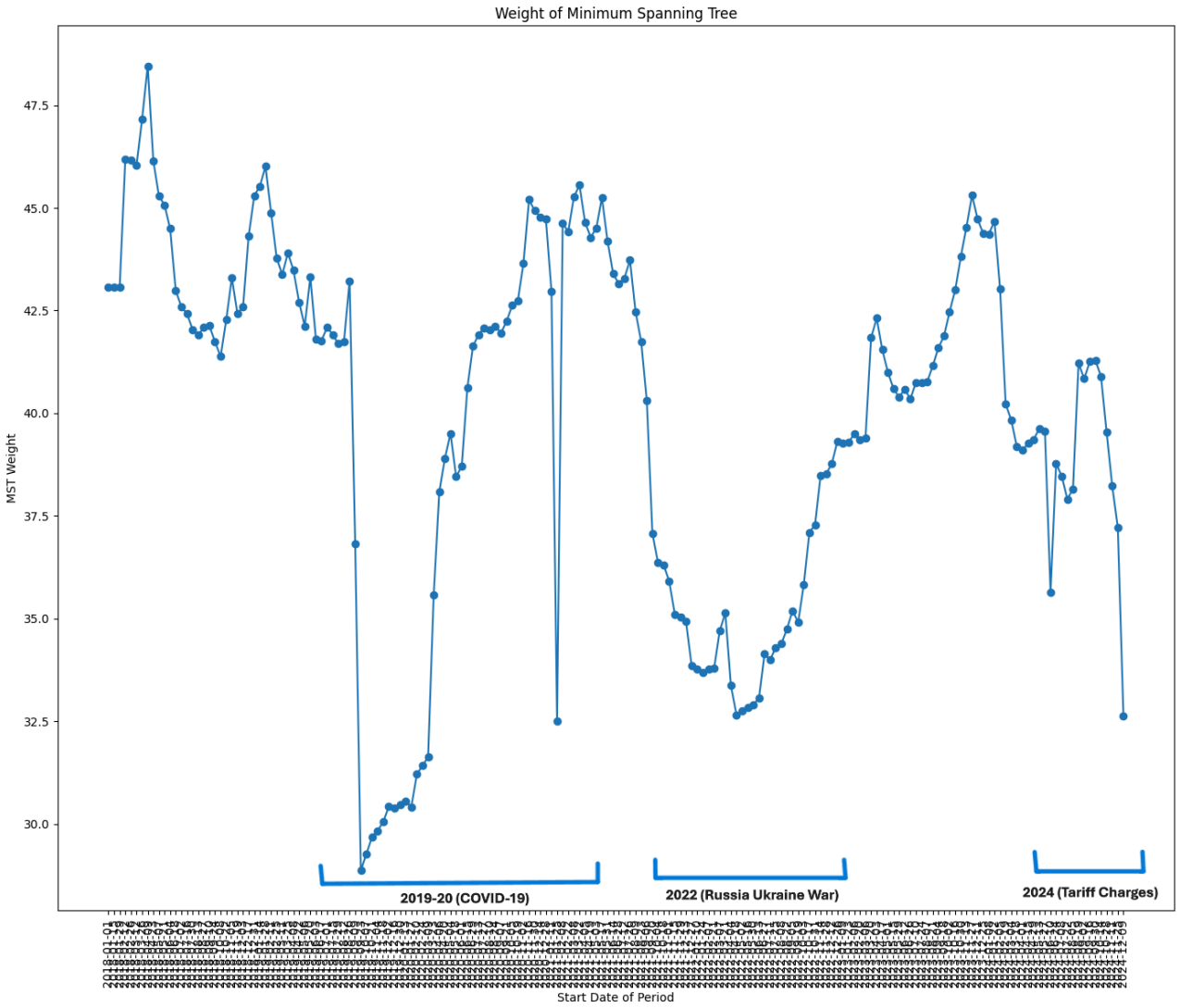
The most influential stocks in both regions were found by the average rank of the five graph centralities as discussed in the methodology.

INDIA	USA
State Bank of India	The Home Depot, Inc.
ICICI Bank Limited	Honeywell International Inc.
Hindalco Industries Limited	Cisco Systems, Inc.
Steel Authority of India Limited	Linde plc
Tata Motors Limited	Microsoft Corporation
JSW Steel Limited	Accenture plc
Grasim Industries Limited	Comcast Corporation
Tata Steel Limited	Danaher Corporation
Axis Bank Limited	Thermo Fisher Scientific Inc.

Mahindra & Mahindra Limited	Morgan Stanley

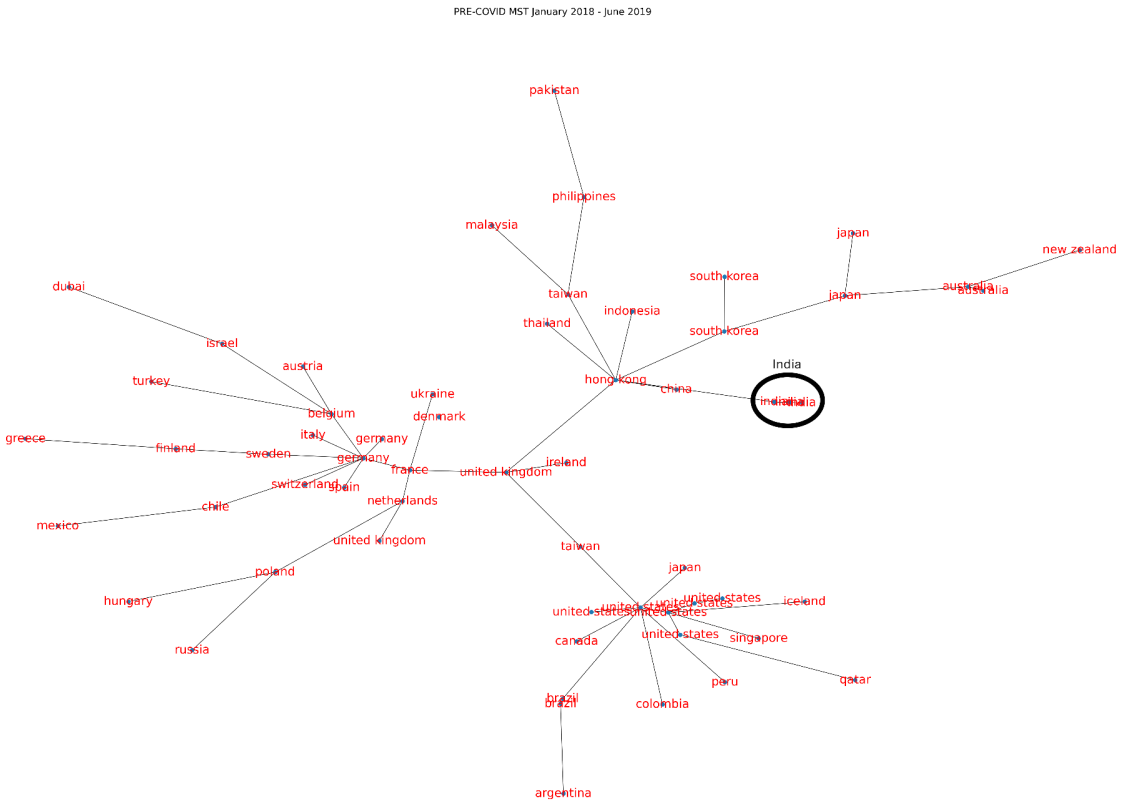
4.4 Global Markets - MST Weight

The time evolution plot of the MST weight showed sudden drops that reflected the increased level of the correlation among global stock indices of different countries. These drops are visible during the time of COVID-19 pandemic (2019-20), beginning of Russian Ukrainian War (2022) and the latter half of 2024 (U.S.-China trade tariff skirmishes).



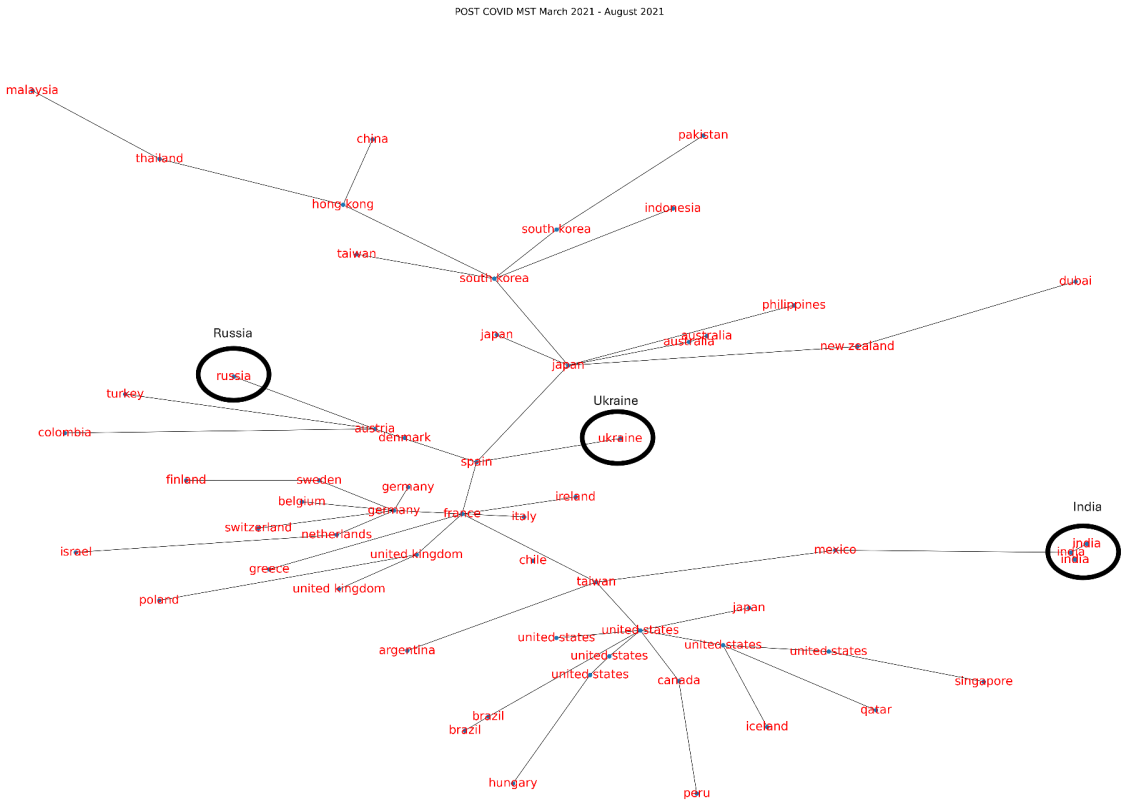
4.5 Migration of Countries from Regional Clusters

MST on different time intervals showed movements of some countries during COVID-19 and the Russian-Ukrainian War as well.



Pre Covid MST

Before the pandemic, India's stock market exhibited stronger ties with Asian markets. But post-COVID (post-COVID MST below), India has migrated towards American and European clusters. This also tells us that during COVID-19, these markets were more hit by the pandemic. The pandemic amplified global financial interconnectedness, with India's National Stock Exchange (NSE) increasingly correlating with U.S. and European indices. During COVID-19, Indian sectors like IT and pharmaceuticals mirrored trends in U.S. tech stocks and European healthcare indices due to shared investor responses to pandemic-driven uncertainties.

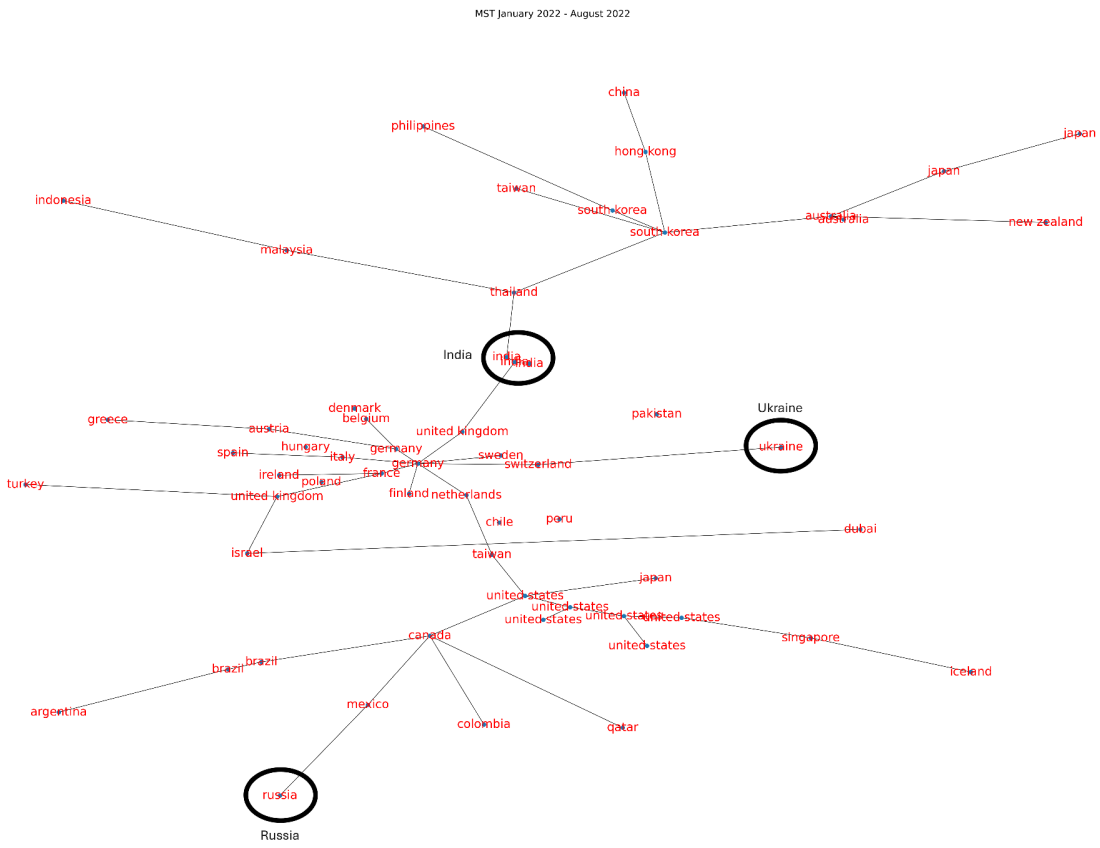


Post Covid MST and Before Russian Ukraine War

Before the start of the Russia-Ukraine War, Russia was the most important trading partner of the EU, as it was the most important supplier of coal, gas, and oil. In return, Russia was dependent on EU imports, which included quantum computing, advanced semiconductors, sensitive machinery, and chemicals. That is the reason Russia had stronger market ties with the European market before the start of the war in 2022.

After Russia invaded Ukraine, the European Commission applied trade restrictive measures in the form of bans on the import and export of goods. It included an import ban on Russian steel products and export bans on luxury goods to Russia. The EU helped Ukraine by supplementing trade concessions and eliminating tariff charges. European markets worked together to redirect Ukraine's export trade via land trade and make available the EU transportation infrastructure.

Moreover, India again started to migrate towards Asian markets away from the American markets as the market started recovering post-COVID-19, and it migrated completely by 2024.



Russia Ukraine War (2022)

5 Conclusion

The thesis demonstrates how techniques like social network analysis and the theory of threshold networks can be applied to study the interrelationships of global financial markets using different network properties and various graph centralities. Using these metrics, stocks in the Indian and U.S. markets were ranked. Topological methods make high-dimensional financial data interpretable and visualizable, helping draw clear inferences and conclusions. The network analysis for the years from 2006 to 2024 shows how an investor can achieve portfolio diversification in Indian markets more easily than in the USA markets due to the strong correlation among stocks in the latter markets. The study also showed the temporal evolution of MST weights of 60 global countries' financial

indices, which captured market crashes during the period of stressful events - COVID-19, the Russia-Ukraine War, and trade tariff skirmishes. Snapshots of Minimum Spanning Trees helped to observe the regional patterns in market trends, combined with the migration of India during the pandemic and Russia and Ukraine during the Russia-Ukraine War.

Together, all these findings tell us that

- MST weight serves as a real-time cohesion metric for global financial markets.
- Graph centralities and network properties accurately reflect the stability of the financial markets.
- Cluster migrations mirror underlying economic, political changes, and policy shifts.

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